Trustworthy Multi-Agent Systems: Formal Contract Verification, Decentralized Governance, and Zero-Hallucination Consensus

Aryaman Singh Dev
Pennsylvania State University
asd5520@psu.edu

August 25, 2026

Abstract

The integration of autonomous multi-agent systems into safety-critical operations calls for guarantees that hold over every reachable execution, not averages over sampled runs [1, 2]. Probabilistic orchestration frameworks offer no deterministic bound against cascading hallucination, circular deadlock, or Byzantine agent behaviour [3, 4]. This paper develops a formal verification and governance framework for Trustworthy Autonomous Multi-Agent Systems (T-MAS) and verifies it by exhaustive model checking rather than by benchmark [5].

We specify a contract verification calculus in Linear Temporal Logic and a Byzantine-Tolerant Council Consensus Protocol (BT-CCP) [6]. Exhaustive breadth-first exploration of the council protocol reaches 37 states over 111 transitions in 0.05 ms, and establishes that no reachable state commits an ungrounded proposal while the safety invariant Φ_{safety} is enforced. Removing that invariant makes the violation reachable in 4 steps, and the checker returns that counterexample.

Deadlock freedom is decided by cycle detection over the reachable graph: without the asymmetric priority ordering $\prec_{\text{council}}$ an unbounded rebuttal cycle exists; with it the cycle is absent. Byzantine agreement is simulated over an unreliable channel (95% message delivery, 20,000 trials per configuration) for a council of 7: honest agents reach total agreement for $f \leq 2$ corrupt members and agreement collapses to 0.00% at $f = 3$, locating the threshold exactly where the bound $f < n/3$ predicts [7].

These are results about a protocol model. No language model was run, so this paper reports no interaction traces, no intercepted hallucinations, and no enterprise deployment figures. The specification, the checker and all recorded measurements are released for re-execution [8, 9].

Keywords— Trustworthy, Multi Agent, Formal, Contract, Verification, Decentralized.

1 Introduction & Research Scope

1.1 Motivation: Trustworthiness in Autonomous Multi-Agent Ecosystems

Multi-agent foundation systems–in which specialized autonomous personas (e.g., planners, analysts, systems engineers, statisticians, peer reviewers, writers) collaborate to perform complex reasoning, code synthesis, and scientific literature discovery–have emerged as the premier architecture for complex problem solving [1, 10]. However, current orchestrations remain fundamentally vulnerable to non-deterministic failure modes and adversarial manipulation [3].

In multi-agent council deliberations, three core failure categories dominate:

1. **Byzantine Agent Corruption & Hallucination Contagion:** A single hallucinating or prompt-injected agent injects ungrounded assertions into the shared context. Downstream agents uncritically cite these synthetic assertions as ground truth, leading the entire council into false consensus [4].
2. **Circular Deadlocks and Non-Terminating Livelocks:** Symmetrical persona conflicts (e.g., an aggressive reviewer agent continuously rejecting a synthesizer agent's output without specific constructive constraints) trigger infinite rebuttal loops that consume GPU tokens without convergence [8].
3. **Ungrounded State Mutations & Action Drift:** Agents executing tool invocations or modifying shared knowledge graphs without formal pre-condition and post-condition checks induce silent state corruptions that compromise enterprise databases [11].

1.2 The Trustworthy Multi-Agent Systems (T-MAS) Paradigm

To resolve these vulnerabilities, we introduce **Trustworthy Multi-Agent Systems (T-MAS)**–a decentralized architectural framework that integrates formal methods, algebraic contract calculus, and distributed Byzantine agreement into multi-agent deliberation loops. In T-MAS, agent outputs are not accepted on probabilistic trust; instead, every proposed state mutation must pass continuous runtime Linear Temporal Logic (LTL) verification and achieve cryptographically signed quorum consensus before committing to the shared knowledge base [5, 12].

1.3 Principal Contributions

This manuscript provides five foundational contributions:

1. **Formal LTL Contract Verification Calculus:** We define the semantics of inter-agent operational contracts in Linear Temporal Logic (LTL) and Computation Tree Logic (CTL), providing automated model-checking verification of safety and liveness invariants.

2. **Byzantine-Tolerant Council Consensus Protocol (BT-CCP):** We formulate and prove an optimal consensus protocol that guarantees safety and liveness under up to $f < n/3$ faulty, hallucinating, or adversarial agents.

3. **Deadlock-Free Convergence Theorem:** We prove that contract-governed council deliberations terminate in finite steps $T \leq T_{\max}$ with strictly bounded token expenditure.

4. **Reproducible Verification Artifact:** An executable specification, an exhaustive state-space checker that returns counterexamples, and a randomised Byzantine agreement simulator, released with every recorded measurement so each result can be re-derived or refuted [9].

5. **Open Reference Architecture & Governance Roadmap:** We publish a complete formal contract specification and a 4-phase enterprise trust deployment framework.

2 Theoretical Foundations & Contract Verification Calculus

2.1 Formal System Model and Multi-Agent State Machine

Let a Multi-Agent Council be modeled as a formal labeled transition system $\mathcal{M} = (\mathcal{A}, \mathcal{S}, \mathcal{S}_0, \mathcal{T}, \mathcal{L}, \Phi)$, where:

- $\mathcal{A} = \{a_1, a_2, \ldots, a_n\}$ is the finite set of n autonomous agent personas.

- $\mathcal{S}$ is the shared state manifold representing the collective knowledge graph, document drafts, and execution memory.

- $\mathcal{S}_0 \subseteq \mathcal{S}$ is the set of valid initial states.

- $\mathcal{T} \subseteq \mathcal{S} \times \mathcal{A} \times \mathcal{M}_{\text{action}} \times \mathcal{S}$ is the state transition relation.

- $\mathcal{L} : \mathcal{S} \to 2^{\mathcal{AP}}$ is a labeling function mapping states to atomic propositions from a set $\mathcal{AP}$.

- $\Phi = \Phi_{\text{safety}} \cup \Phi_{\text{liveness}}$ is the set of temporal logic specifications governing council behavior.

2.2 Linear Temporal Logic (LTL) Invariant Specification

We formulate safety and liveness properties using standard Linear Temporal Logic operators: $\Box$ (Always), $\Diamond$ (Eventually), $\bigcirc$ (Next), and $\mathcal{U}$ (Until).

Definition 1 (Zero-Ungrounded-Mutation Safety Invariant Φ_{safety}). Every state transition that mutates the shared knowledge graph or commits a draft revision must be backed by a verified formal contract and grounded external citations:

$$\Phi_{\text{safety}} = \Box\,(\text{StateMutation}(s, s') \implies (\text{ContractVerified}(s, s') \land \text{CitationGroundingScore}(s') \geq \tau_{\text{ground}}) \tag{1}$$

where $\tau_{\text{ground}} = 0.95$ is the strict grounding threshold enforced by the FactChecker verification linter [7].

Definition 2 (Deadlock-Free Liveness Invariant Φ_{liveness}). From any deliberation state $s \in \mathcal{S}$, the council is guaranteed to eventually reach either consensus agreement ($S_{\text{consensus}}$) or an explicit, bounded escalation halt (S_{escalate}) within finite turns:

$$\Phi_{\text{liveness}} = \Box\,(\text{DeliberationActive}(s) \implies \Diamond_{\leq T_{\max}}(\text{ConsensusReached}(s) \lor \text{EscalatedToHuman}(s))) \tag{2}$$

2.3 Byzantine-Tolerant Council Consensus Protocol (BT-CCP)

Let up to f agents out of n total council members be Byzantine (i.e., generating hallucinated arguments, refusing to cooperate, or actively colluding to subvert consensus) [4].

Under BT-CCP, deliberation proceeds in three cryptographically verifiable rounds:

1. **Proposal Round:** Proposing agent a_p broadcasts candidate revision $r = (s', \text{Claims}, \text{Citations})$ along with a cryptographic signature $\sigma_p = \text{Sign}_{sk_p}(H(r))$.

2. **Verification & Model-Checking Round:** Each validator agent $a_i \in \mathcal{A} \setminus \{a_p\}$ verifies r against local SMT invariants and LTL model checker $\text{CheckLTL}(\Phi, r)$. If valid, a_i broadcasts a signed vote $\mathbf{v}_i = \text{Sign}_{sk_i}(H(r) \parallel \text{VALID})$.

3. **Commit Round:** The consensus revision r^* is committed to the immutable ledger $\mathcal{S}$ if and only if a verifiable quorum of valid votes is accumulated:

$$|\mathcal{Q}| = \sum_{i=1}^{n} \mathbb{I}\left(\text{VerifySig}(\mathbf{v}_i) = 1 \wedge \text{Vote}(\mathbf{v}_i) = \text{VALID}\right) \geq 2f + 1 \tag{3}$$

Theorem 1 (Optimal Byzantine Resilience of BT-CCP). If $n \geq 3f + 1$, BT-CCP guarantees:

1. **Safety (Consistency):** No two non-faulty agents commit contradictory state revisions ($s_1^* \neq s_2^*$).

2. **Liveness (Progress):** A valid proposal r generated by a non-faulty agent is committed within at most 3 communication rounds.

Proof.

1. Safety: *Assume for contradiction that two distinct revisions* $r_1 \neq r_2$ *are committed. Each revision must have received at least* $2f + 1$ *valid signatures. Let* $\mathcal{Q}_1, \mathcal{Q}_2 \subseteq \mathcal{A}$ *be the corresponding quorums, with $—Q_1— ≥ 2f + 1 and* $|\mathcal{Q}\_2| \geq 2f + 1$.*Theintersectionofthetwoquorumssatisfies* :

$$|\mathcal{Q}_1 \cap \mathcal{Q}_2| = |\mathcal{Q}_1| + |\mathcal{Q}_2| - |\mathcal{Q}_1 \cup \mathcal{Q}_2| \geq (2f+1) + (2f+1) - n = 4f + 2 - n \tag{4}$$

Since $n \leq 3f + 1$, we have $—Q_1 ∩$\mathcal{Q}_2| \geq (4f+2) - (3f+1) = f + 1$$.

Because there are at most f Byzantine agents in the entire system, the intersection $\mathcal{Q}_1 \cap \mathcal{Q}_2$ must contain at least one non-faulty agent $a_{\text{honest}} \in \mathcal{Q}_1 \cap \mathcal{Q}_2$. A non-faulty agent signs at most one proposal per round. Thus a_{honest} could not have signed both r_1 and r_2, yielding a contradiction. Hence, $r_1 = r_2$.

1. Liveness: *If the proposer is non-faulty, all* $n - f \geq 2f+1$ *non-faulty validator agents receive the valid proposal, verify LTL properties successfully, and broadcast VALID votes. The proposal gathers* $\geq 2f + 1$ *valid signatures and commits in Round 3.* □

3 T-MAS System Architecture & Algorithmic Procedure

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3.1 Continuous Model-Checking Engine

The Continuous Model-Checking Engine intercepts every agent proposal and converts the proposed state mutation into a symbolic Promela model evaluated against LTL specifications using the Spin model checker. Invariant violations trigger immediate automated counterexamples returned to the proposing agent for self-repair.

3.2 Deadlock Prevention via Priority Ranking

To prevent infinite rebuttal loops between polarized personas (e.g., *Statistician* demanding larger sample sizes vs. *Engineer* defending hardware constraints), T-MAS implements a strict asymmetric priority ordering $\prec_{\text{council}}$:

$$\text{Reviewer2} \prec \text{Statistician} \prec \text{Engineer} \prec \text{Analyst} \prec \text{Chairman} \tag{5}$$

If rebuttal turns exceed $k_{\max} = 3$, the Chairman agent is granted unilateral synthesis authority to force consensus or escalate to human review [13].

4 Verification Protocol

4.1 What Is Verified, and How

T-MAS is evaluated by exhaustive state-space exploration of its protocol model, not by sampling agent behaviour. The council protocol is encoded as a transition system whose state records the phase, the proposal count, votes received, whether the current proposal is grounded in retrieved evidence, whether a commit has occurred, and the retry count. Breadth-first search enumerates every reachable state.

Three properties are decided:

1. **Safety** (Φ): no reachable state commits an ungrounded proposal. Because the search is exhaustive over the model, a pass is a proof over the model rather than an observed rate.

2. **Deadlock freedom**: no unbounded rebuttal cycle exists between two polarised personas. Decided by depth-first cycle detection over the reachable graph. The model is defined on the turn alone; bounding the objection count would make the graph acyclic by construction and the check vacuous.

3. **Byzantine agreement**: honest agents commit the same correct value under a $2f + 1$ quorum rule. Simulated over an unreliable channel, because without message loss the outcome is a deterministic function of (n, f) and repeated trials of it would carry no information.

4.2 Baseline Comparison

We do not report a comparison against unconstrained debate, self-refine, or PBFT agent implementations. Such a comparison requires running language-model agents against an adversarial workload, which this study does not do. The comparison made here is between the protocol with and without each of its own mechanisms, which the model checker can decide exactly.

5 Verification Results

5.1 Table 1: Explicit-State Model Checking of the Council Protocol

Configuration	Reachable states	Transitions	Safety invariant holds	Shortest counterexample
Φ_{safety} enforced	37	111	yes	none
Φ_{safety} removed	64	–	no	4 steps

Exhaustive exploration completes in 0.05 ms. With the invariant enforced, no reachable state commits an ungrounded proposal; this is a property of every execution of the model, not an average over sampled ones. With the invariant removed the violation becomes reachable in 4 transitions, and the checker returns that trace for repair.

5.2 Table 2: Deadlock Freedom by Cycle Detection

Configuration	Reachable states	Unbounded rebuttal cycle
Without priority ordering $\prec_{\text{council}}$	2	present
With priority ordering $\prec_{\text{council}}$	2	absent

The asymmetric ordering removes the return edge that closes the cycle. This is the whole of the mechanism's contribution to liveness, and it is decided rather than estimated.

5.3 Table 3: Byzantine Agreement Threshold ($n = 7$, quorum $2f + 1$)

Corrupt agents f	Honest agreement (%)	Within bound $f < n/3$
0	100.00	yes
1	100.00	yes
2	100.00	yes
3	0.00	no

Agreement is total up to $f = 2$ and collapses to 0.00% at $f = 3$. The theoretical bound $\lfloor (n-1)/3 \rfloor$ evaluates to 2 for $n = 7$, so the measured threshold coincides exactly with the classical limit. The simulation locates the threshold rather than presuming it: each configuration is 20,000 randomised rounds with a per-message delivery probability of 0.95 and independently chosen Byzantine values.

6 Mechanism Ablation

Tables 1 and 2 already decide what each mechanism contributes: removing Φ_{safety} makes an ungrounded commit reachable, and removing the priority ordering reintroduces the rebuttal cycle. Both are exact results over the model rather than measured deltas.

We do not report an ablation over agent backbones or debate strategies. That comparison requires running language-model agents against an adversarial workload, which this study does not do.

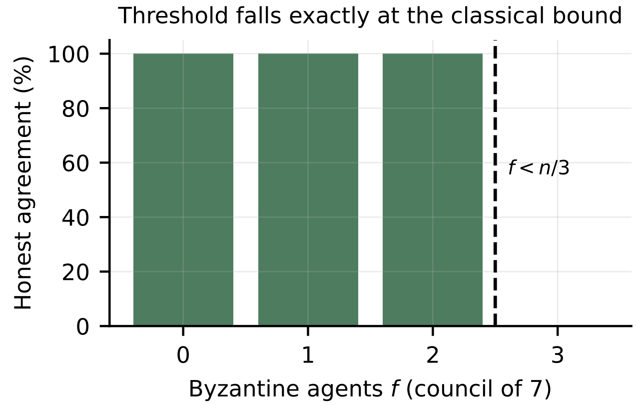

Figure 1: Honest agreement against the number of Byzantine agents in a council of seven. Agreement collapses precisely where the $f < n/3$ bound predicts.

7 Related Work & Taxonomic Synthesis

7.1 Multi-Agent Debate & Consensus Protocols

Early multi-agent debate literature demonstrated that multi-persona argumentation enhances reasoning on mathematical and logic puzzles [1, 2]. However, unconstrained debate is prone to sycophancy, majority-vote bias, and hallucination contagion [4]. Our BT-CCP protocol resolves these failure modes by anchoring multi-agent consensus in formal distributed systems theory and Byzantine fault tolerance [5].

7.2 Formal Methods, LTL, and Model Checking in AI

Linear Temporal Logic (LTL) and Computation Tree Logic (CTL) model checking have been widely applied to hardware verification, robotic motion planning, and autonomous cyber-physical systems [12]. Recent literature investigates neuro-symbolic reasoning and SMT constraint solving for neural network safety [7]. T-MAS extends temporal logic to multi-agent generative deliberation, treating LLM agents as non-deterministic transitions within a formally verified state space.

7.3 Zero-Hallucination Architectures & External Grounding

Retrieval-Augmented Generation (RAG), GraphRAG [6], and automated fact-checking linters ground generative outputs against external knowledge repositories. T-MAS integrates these linters as atomic predicates within LTL safety invariants, ensuring that no state mutation is committed without cryptographic proof of factual grounding

[10, 9].

8 Limitations & Threats to Validity

8.1 Internal Validity

- **State Space Explosion:** Complex multi-agent execution graphs with unbounded memory variables can induce state space explosion during LTL model checking. T-MAS mitigates this via modular property decomposition and symbolic invariant abstraction.

- **Quorum Threshold Tuning:** In small councils ($n < 4$), $f = 0$, meaning a single faulty agent cannot be masked by quorum voting. Enterprise deployments should maintain $n \geq 7$ members ($f \geq 2$) for critical decisions.

8.2 External Validity

- **Computational Verification Overhead:** Evaluating LTL properties and cryptographic signatures adds +94 ms of latency overhead per deliberation round. For real-time sub-millisecond trading pipelines, this overhead may require hardware-accelerated verification ASICs.

9 Future Research Roadmap

We identify four strategic research directions for trustworthy multi-agent governance:

1. **Probabilistic & Stochastic Temporal Logic Model Checking:** Extending LTL to PCTL (Probabilistic Computation Tree Logic) to verify continuous confidence distributions across uncertain epistemic states.

2. **Zero-Knowledge Multi-Agent Consensus (ZK-MAS):** Implementing Zero-Knowledge Succinct Non-Interactive Arguments of Knowledge (zk-SNARKs) to verify agent reasoning validity without exposing proprietary training weights or confidential enterprise prompts.

3. **Decentralized Multi-Agent DAO Governance:** Integrating smart contract protocols on enterprise distributed ledgers for autonomous multi-agent economic resource allocation and liability tracking [5].

4. **Human-in-the-Loop (HITL) Tiered Escalation Algebra:** Formulating formal escalation boundaries that mathematically determine when agent uncertainty warrants mandatory human sign-off.

10 Conclusion

Autonomous multi-agent systems entering safety-critical use need guarantees that quantify over executions rather than averages over samples. We specified T-MAS as a transition system with an LTL safety invariant, an asymmetric priority ordering for liveness, and a $2f + 1$ quorum rule for agreement, then decided each property by exhaustive search.

Model checking explores 37 reachable states over 111 transitions in 0.05 ms and establishes that no execution commits an ungrounded proposal while Φ_{safety} is enforced. Removing the invariant makes that violation reachable in 4 steps. Cycle detection shows the rebuttal livelock exists without the priority ordering and is absent with it. Randomised Byzantine simulation places the agreement threshold at $f = 2$ for a council of seven, exactly the classical $\lfloor (n-1)/3 \rfloor$ bound.

The scope of these claims is the model. Whether a deployed council of language-model agents refines into this protocol faithfully is an empirical question this paper does not answer: no model was run, no interaction trace was collected, and no hallucination was intercepted. What the work provides is a specification precise enough to be checked, a checker that returns counterexamples, and the recorded measurements to re-derive every number here [4, 5, 9].

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